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Exercise 4: Launching Web Applications using App Engine Tools

Aim

To launch and run a web application locally using Google App Engine tools (Flask) and verify it in a browser.

Procedure

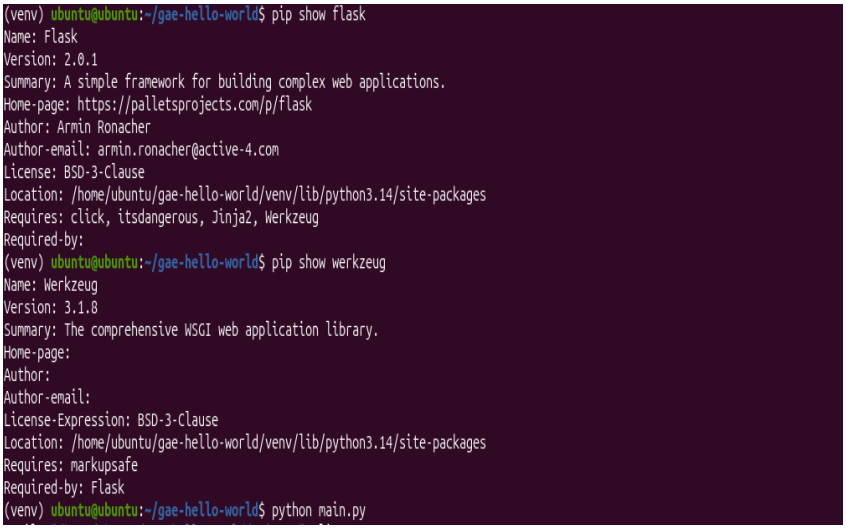
Step 1: Upgrade Flask and Werkzeug

```
pip install --upgrade flask werkzeug
```

This upgrades Flask and its dependencies (Werkzeug, blinker, etc.) in the virtual environment.

Step 2: Verify Installed Package Versions

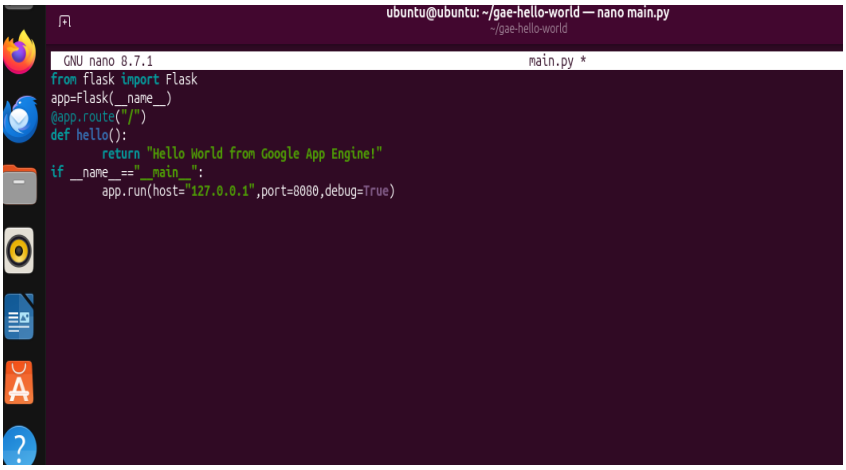
```
pip show flask
pip show werkzeug
```



Step 3: Review Application Code (main.py)

main.py defines a simple Flask route that returns a Hello World message:

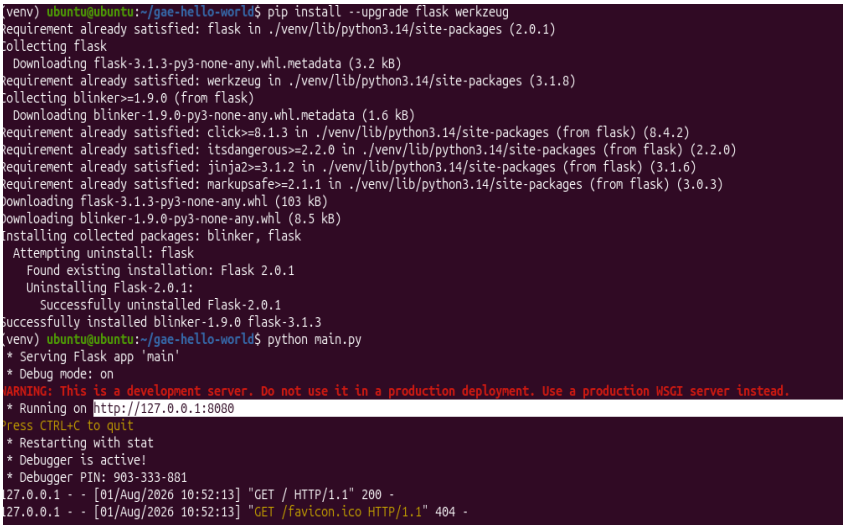
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Step 4: Run the Application

`python main.py`

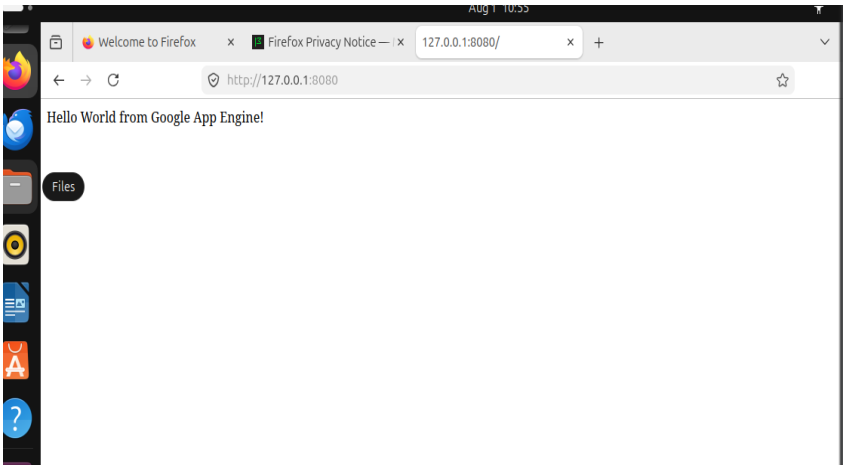
Flask starts the development server on http://127.0.0.1:8080 with debug mode enabled.



Output

The application was accessed from the browser at 127.0.0.1:8080, confirming successful deployment:

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Result

Thus, the Flask-based web application was successfully launched and tested locally using Google App Engine tools.